

2024



GovTech Collaborates with AWS Generative AI Innovation Center to Scale Generative AI Adoption Across Public Sector

Learn how the statutory board responsible for delivering digital services to the public in Singapore built an end-to-end AI/ML development platform

[Overview](#) | [Opportunity](#) | [Solution](#) | [Our](#)



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



75%

Improved cost-performance for generative AI workloads

No-Code ML Interface

Simplifies the building, training, and deploying of ML models

Production-Ready Generative AI

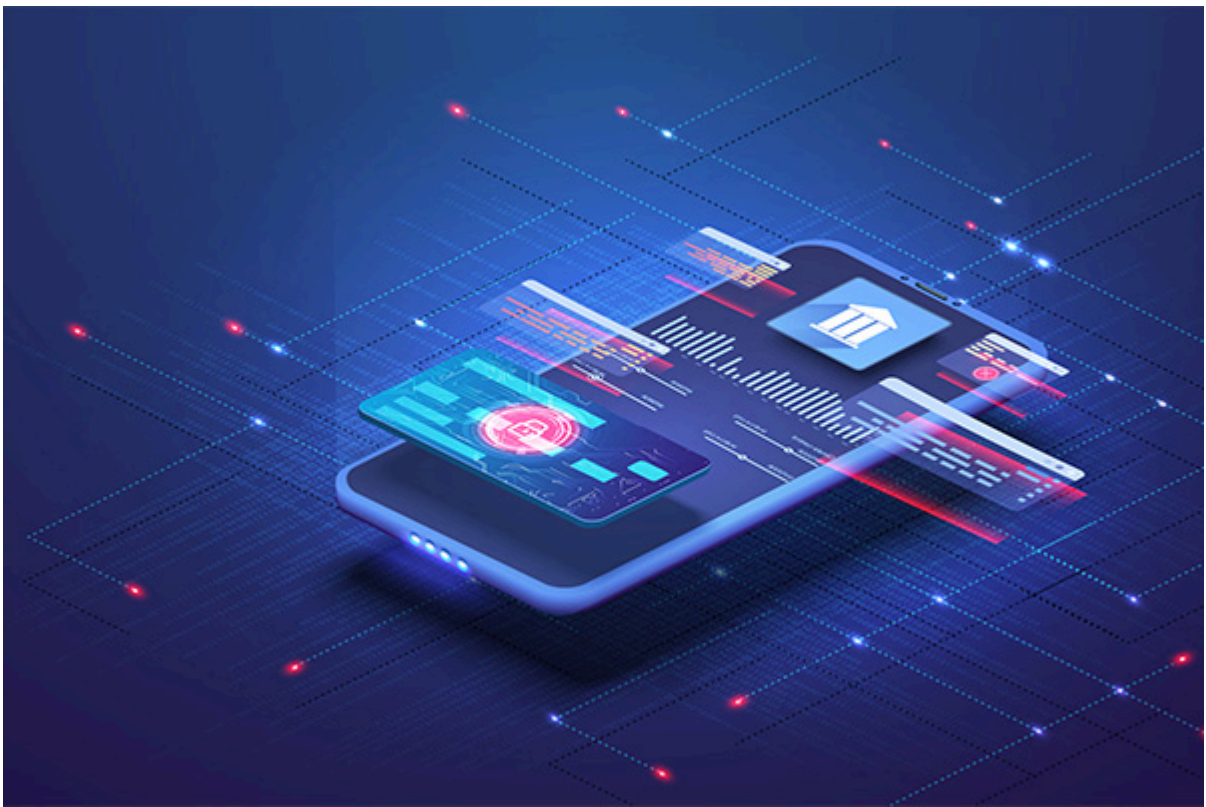
Fast-tracks use cases in agencies across the Singapore Government

Overview

From streamlining administrative processes and reducing wait times to personalizing interactions, generative artificial intelligence (generative AI) has the potential to transform public service delivery with tailored, purpose-built solutions that better meet the community's needs.

The [Government Technology Agency](#) (GovTech) is a statutory board of the Government of Singapore responsible for the delivery of digital services to the public. As the Centre of Excellence for Infocomm Technology and Smart Systems, GovTech also develops the Singapore government's capabilities in data science, including generative AI.

In 2023, GovTech collaborated with the [AWS Generative AI Innovation Center](#) at [Amazon Web Services](#) (AWS) to integrate generative AI capabilities into **MAESTRO** (abbreviation for **M**achine Learning & **AI** Enterprise-level **S**ecure **T**ool-Suite for **R**eliable **O**perations). MAESTRO is an end-to-end artificial intelligence (AI) and machine learning (ML) development solution, which allows public sector organizations to perform cost-efficient AI/ML operations. Since its launch in August 2023, agencies like the Ministry of Manpower (MOM) and the Central Provident Fund Board (CPF Board) used MAESTRO to develop generative AI-powered tools that enhanced their ability to deliver citizen services.



Opportunity | Overcoming High Cost Demands for Generative AI Operations

GovTech identified generative AI as a way to drive innovation, enhance public services, and improve lives within the community. However, the high costs of operating LLMs at a government scale were key considerations for the organization.

To address cost concerns and tap into the potential of generative AI, GovTech collaborated with the AWS Generative AI Innovation Center to develop generative AI solutions that enable cost-effective adoption of this transformative technology across government agencies.

To build and deploy MAESTRO, GovTech worked with AWS, including the AWS Generative AI Innovation Center, a program that pairs organizations with AWS science and strategy experts with deep experience in AI/ML and generative AI techniques. Public sector organizations can now use MAESTRO, to create cost-efficient generative AI tools tailored to agency-specific use cases. Within 9 months of launch, MAESTRO was adopted by 20 public sector organizations, corresponding to over 45 project teams and over 300 data scientists and ML engineers across Singapore.

MOM, the ministry responsible for the formulation and implementation of workforce policies in Singapore, used MAESTRO to drive quicker operations and policy interventions. It developed a generative AI sensemaker tool that uses clustering algorithms and an LLM to extract insights from documents. Within 3 months, MOM used the tool to process over 1 million documents, improve insights extraction by 60 percent, reduce sensemaking time by 50 percent, and save over 2,000 work hours.

MOM also enhanced job matching for employees and job seekers. It developed the Singapore Standard Occupational Classification (SSOC) Autocoder, which automatically sorted job postings based on occupational codes. Within 3 months following its launch, the SSOC Autocoder had processed 10 million job postings with 92 percent accuracy.

The Central Provident Fund Board (CPFB) of Singapore is a statutory board whose mission is to enable Singaporeans to have a secure retirement through lifelong income, healthcare financing, and home financing. CPFB used open-source, quantized Generative AI models deployed through MAESTRO to summarize call transcripts for the ~600,000 calls that it receives on average each year. These summaries are vital for follow-ups, investigations, and quickly identifying emerging issues.

GovTech MAESTRO incorporated generative AI capabilities using [Amazon Bedrock](#), a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies, and [Amazon SageMaker JumpStart](#), an ML hub with foundation models, built-in algorithms, and pre-built ML solutions. With ready-made models, public sector organizations can build and deploy AI solutions without investing significant time and resources in training models from scratch.

Furthermore, with quantization techniques, teams can deploy multiple smaller, right-sized models instead of a single large general model. Overall, by working with the AWS Generative AI Innovation Center, MAESTRO improved cost performance for generative AI workloads by up to 75 percent.

To simplify the creation, collaboration, and management of ML workflows, GovTech built MAESTRO using [Amazon SageMaker Studio](#), a single web-based interface for end-to-end ML development, and [Amazon SageMaker Canvas](#), a no-code ML interface. With a unified, web-based, no-code interface, MAESTRO reduces complexity and simplifies the building, training, and deploying of ML models even for non-technical staff. By lowering the barriers to entry, MAESTRO has enabled government agencies across the board to rapidly adopt and leverage cost-effective, production-ready Generative AI solutions, accelerating the implementation of various use cases.

"Leading organizations realize that scaling generative AI cost-effectively is critical. We're proud to collaborate with GovTech to make generative AI production-ready across Singapore's public sector," said Nieves Garcia, AWS Generative AI Innovation Center Strategy Lead for Asia Pacific, Japan and Greater China Region.

Outcome | Expanding Generative AI Adoption Across the Public Sector



GovTech plans to deepen its partnership with AWS to expand generative AI and AI/ML operations and best practices to more government agencies. By optimizing for cost efficiency, GovTech aims to facilitate fiscally responsible generative AI development among public sector organizations.

“We're thrilled to partner with AWS to make generative AI more accessible and sustainable for our agencies. This allows us to harness generative AI's potential while responsibly managing resources,” said Jeffrey Chai, Product Manager of MAESTRO, Government Technology Agency.

Learn More

To learn more, visit aws.amazon.com/government-education/government.

About Government Technology Agency

The [Government Technology Agency](#) (GovTech) is a statutory board of the Government of Singapore responsible for the delivery of digital services to the public. As the Centre of Excellence for Infocomm Technology and Smart Systems, GovTech is also responsible for developing the Singapore government's capabilities in data science, artificial intelligence, application development, smart city technology, digital infrastructure, and cybersecurity.

AWS Services Used

Amazon SageMaker

Amazon SageMaker is a fully managed service that brings together a broad set of tools to enable high-performance, low-cost machine learning (ML) for any use case.

[Learn more »](#)

Amazon SageMaker Studio

Amazon SageMaker Studio offers a wide choice of purpose-built tools to perform all machine learning (ML) development steps, from preparing data to building, training, deploying, and managing your ML models.

[Learn more »](#)



Amazon SageMaker Canvas

SageMaker Canvas provides access to ready-to-use models including foundation models from Amazon Bedrock or Amazon SageMaker JumpStart or you can build your own custom ML model using AutoML powered by SageMaker AutoPilot.

[Learn more »](#)

Amazon SageMaker JumpStart

Amazon SageMaker JumpStart is a machine learning (ML) hub that can help you accelerate your ML journey. With SageMaker JumpStart, you can evaluate, compare, and select FMs quickly based on pre-defined quality and responsibility metrics to perform tasks like article summarization and image generation.

[Learn more »](#)

More Public Sector Customer Stories

GERMANY



HeyJobs on AWS

HeyJobs aims to make job searching and recruitment more effective for candidates and employers.

2024

ITALY



Namirial Streamlines Contract Signing with...

Namirial enhanced its eSignAnywhere solution with generative AI using Amazon Bedrock, improving engineering productivity and the user



2024

COSTA RICA

AUSTRALIA





Modernizing Applications in Minutes Using Amazon Q...

Learn how Novacomp accelerated



2024



Dovetail Enhances Customer Insights with...

Learn how Dovetail accelerates development of new generative AI products and features using Amazon Bedrock, deploying the latest large language models while reducing security and administrative overhead.

2024

1 2 3 4 5 6 ... 201

Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering on their missions every day using AWS. Contact our experts and start your own AWS journey today.

Contact Sales

Resources for AWS

Getting Started
Training and Certification
AWS Solutions Library
Architecture Center
Product and Technical FAQs
Analyst Reports
AWS Partners

Developers on AWS

Developer Center
SDKs & Tools
.NET on AWS
Python on AWS
Java on AWS
PHP on AWS
JavaScript on AWS

Help

Contact Us
Get Expert Help
File a Support Ticket
AWS re:Post
Knowledge Center
AWS Support Overview
Legal
AWS Careers

Learn About AWS

What Is AWS?
What Is Cloud Computing?
AWS Accessibility
AWS Inclusion, Diversity & Equity
What Is DevOps?
What Is a Container?
What Is a Data Lake?
What is Artificial Intelligence (AI)?
What is Generative AI?
What is Machine Learning (ML)?





Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

Language

- [عربي |](#)
- [Bahasa Indonesia |](#)
- [Deutsch |](#)
- [English |](#)
- [Español |](#)
- [Français |](#)
- [Italiano |](#)
- [Português |](#)
- [Tiếng Việt |](#)
- [Türkçe |](#)
- [Русский |](#)
- [ไทย |](#)
- [日本語 |](#)
- [한국어 |](#)
- [中文 \(简体\) |](#)
- [中文 \(繁體\)](#)

Privacy

- [|](#)
- [Accessibility](#)

- [|](#)
- [Site Terms](#)

- [|](#)
- [Cookie Preferences](#)